

Space Missions and Systems

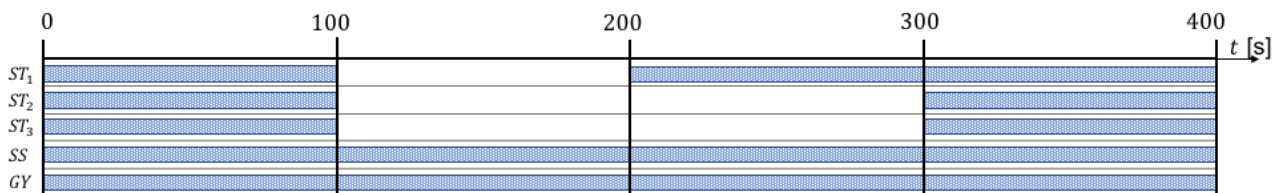
Homework 2

The Mercury Planetary Orbiter (MPO) of the BepiColombo mission will survey the planet Mercury to study its surface, interior, magnetic field, and exosphere. The attitude determination system of the probe consists of a set of three star trackers (ST), a Sun sensor (SS) and an inertial measurement unit (IMU) with gyroscopes (GY).

The MPO will be in a polar 480x1500 km orbit. During science operations, the MPO will maintain nadir-pointing to observe the surface of Mercury. The task of this homework is to compute the MPO attitude by using the data reported in the matrices **OBS**, which include STs and SS measurements, and **OMEGA_MEAS**, which stores the spacecraft angular velocity measured by the gyroscopes. Regarding the gyroscopes error model consider the values of $\sigma_v = 10^{-7} \text{ rad/s}$ and $\sigma_u = 5 \times 10^{-9} \text{ rad/s}^2$ (**Note:** σ_v is the standard deviation of η_1 , and its unit is rad/s, whereas σ_u is the standard deviation of η_2 that is the RRW, $\dot{b} = \eta_2$. Therefore, to include σ_u in the initial *a priori* covariance matrix, P_0 , and in the process noise $\mathbf{Q}$, you need to account for the delta time with respect to the **latest correction** of the bias).

For the first 100 s, the STs work properly, and all measurements are available. After 100 s, the star trackers have a failure, and only GY and SS data are available. At 200 s, the first ST is back on, and after 300 s the suite of STs is on.

Here below is reported the measurements availability over the 400 s.



The **data.mat** file contains the following input data

Name	Size	Structure
Qtrue	4000x5	Simulated quaternions that represent the “true values”. The first 3 columns are the vector part of the quaternion $\vec{q}$, the fourth column is the scalar part of the quaternion, and the fifth column is the observable count time from the initial epoch ($t_0=0$).
OMEGA_TRUE	4000x3	Simulated angular rates that represent the “true values” without any gyroscope errors.
OMEGA_MEAS	4000x3	Measured angular rates that include the error of the gyros.
OBS	10x7x4000	A 3-D matrix that include all ST and SS observations. The first 9 rows store the three different measurements collected by each star tracker (in order ST_1, ST_2, ST_3) while the 10 th row stores the measurements collected by the SS, at the same time t_i . The 7 columns are W_i , V_i , and σ_i for each measurement. The time evolution is stored in the 3 rd dimension of the matrix.

		Note: <u>If the instrument is not collecting a measurement, the corresponding row of the matrix OBS is made by all zeros.</u>
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Task:

With the provided input data

- 1) Provide an estimate of the attitude of the spacecraft over the entire time span $t = [0s, 400s]$.
- 2) Comment and analyze the results obtained during each observation phase.
- 3) Determine what is the attitude determination algorithm that enables the highest performances by plotting the retrieved error and uncertainty on the quaternion and on the angular velocity and the estimate of the gyroscope bias.

Additional information:

The solution of the homework is considered valid if the attitude determination algorithms are implemented by the student, and not based on functions available in the MATLAB toolbox.

For example, the Wahba problem cannot be solved by using the MATLAB function `eig/eigs` or the Kalman filter by using the MATLAB function `kalman`.

However, you can use those functions to test your results. MATLAB functions that are used for matrix and vector operations (e.g., matrix inversion, `inv`) are allowed.

Upload a working computer code (MATLAB or any other programming language) and a short, but sufficiently explanatory, note (a **pdf** file) with the mathematical procedure, results, comments, and figures on the Google Classroom of the course.

Should you have problems in uploading the files on Google Classroom send the files by email to antonio.genova@uniroma1.it **AND** gael.cascioli@uniroma1.it **AND** paolo.cappuccio@uniroma1.it.

All questions must be sent to the same email addresses. Questions addressed to only one of aforementioned three emails will not be answered.

The deadline is **17 May 23:59 CEST**.

Be concise and go straight to the point.

In the note, indicate your **first name**, **last name**, and student id (a.k.a., “numero di matricola”).